

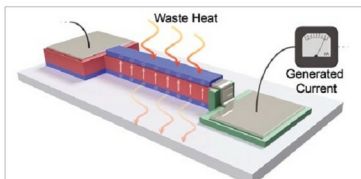
NASA to put robotic bees on Mars

NASA has teams of researchers working on designing a robotic bee that can fly on planet Mars. The cyber-insects, named Marsbees, each about the size of a bumblebee, will have various sensors and huge, battery-powered wings for generating enough lift to hover in the Martian atmosphere, which is around 100 times thinner than Earth's.

Developed by scientists from the US and Japan, the robotic bees will be fitted with sensors and wireless communication devices to map terrain, take samples and look for signs of life, such as methane emissions. The bees will launch from a mobile base that acts as a communications hub and recharging station. These creatures could potentially explore parts of Mars that a land-based rover could never reach.

Thin film converts heat from electronics into energy

Nearly 70 per cent of the energy produced in the US each year is wasted as heat. Engineers at University of California, Berkeley, have developed a thin-film system that can be applied to sources of waste heat—such as computers, cars or large industrial processes—to produce energy at levels unprecedented for this kind of technology.



The illustration of the thin-film device shows how the system converts waste heat into energy (Credit: Shishir Pandya, University of California, Berkeley)

The thin-film system uses a process called pyroelectric energy conversion, which the study demonstrates is well-suited for tapping into waste heat energy supplies below 100°C, called low-quality waste heat. Pyroelectric energy conversion, like many systems that turn heat into energy, works best using thermodynamic cycles (kind of like how a car engine works). But unlike the engine in your car, pyroelectric energy conversion can be realised entirely in the solid state with no moving parts as it turns waste heat into electricity.

The results suggest that this nano-scopic thin-film technology might be particularly attractive for installing on and harvesting waste heat from high-speed electronics but could have a large scope of applications.

NASA, Lockheed Martin set to build quieter supersonic planes

NASA has awarded Lockheed Martin a contract for the design, building and testing of a supersonic aircraft that reduces a

sonic boom to a gentle thump, in order to reintroduce supersonic flight.

Lockheed Martin Aeronautics Co. of Palmdale, California, was selected for the low-boom flight demonstration contract, a cost-plus-incentive-fee contract, valued at US\$ 247.5 million. Work under the contract began on April 2 and will run through December 31, 2021.

Under the contract, Lockheed Martin will complete the design and fabrication of an experimental aircraft, known as X-plane, which will cruise at 16.8km (55,000-feet) at a speed of about 940mph and create a sound at 75 perceived level decibels (PLdB)—about as loud as a car door closing—instead of a sonic boom.

Once NASA accepts the aircraft from the contractor in late 2021, the agency will perform additional flight tests to prove the quiet supersonic technology works as designed, aircraft performance is robust and it is safe to operate in National Airspace System.

Beginning in mid-2022, NASA will fly X-plane over select US cities and collect data about community responses to the flights. This data set will be provided to American and international regulators for their use in considering new sound-based rules regarding supersonic flight over land. This could enable new commercial cargo and passenger markets in faster-than-sound air travel.

Truly wearable smart tech is here

Researchers at University of Manchester, the UK, and Shandong University in China have together developed an ultrafast, nanoscale transistor, commonly known as a thin-film transistor, or TFT, which is made from an oxide semiconductor. This is the first oxide-semiconductor-based transistor capable of operating at a benchmark speed of 1GHz. This could potentially make the next-generation electronic gadgets faster, brighter and more flexible than ever before.

Current TFTs are silicon-based, which makes these opaque, rigid and expensive in comparison to the oxide semiconductor family of transistors that the team is developing. While oxide TFTs will improve pictures on LCD displays, it is their flexibility that is impressive.

According to Aimin Song, professor of nanoelectronics at School of Electrical & Electronic Engineering, University of Manchester, televisions can already be made extremely thin and bright. This work may help make these more mechanically-flexible and cheaper to produce. Also, the GHz transistors may enable medium- or high-performance flexible electronic circuits, such as truly wearable electronics.

Kirigami set to improve smart clothing

In a study published in the journal *Advanced Materials*, a University at Buffalo-led research team describes how Kirigami has inspired efforts to build malleable electronic circuits. Their innovation involves creating tiny sheets of strong yet bendable electronic materials



Shengqiang Ren
(Credit: www.buffalo.edu)